

Chapter 4

4.1 Field Effect Transistor

The FET (Field Effect Transistor) is a three-terminal electronic device used to control the flow of current by the voltage applied to its gate terminal. The three terminals in this device are named drain, source, and gate. Figure 4.1 shows the construction of a FET.

- **Source:** It is a terminal through which charge carriers enter the channel.
- **Drain:** It is a terminal through which charge carriers leave the channel.
- **Gate:** This terminal controls the conductivity between the source and the drain terminals.

FETs are also known as unipolar transistors as opposed to BJTs that are bipolar transistors. In FETs, either holes or electrons are used for the conduction process. Whereas, conduction process doesn't involve both charge carriers simultaneously. FETs usually come with high input impedance at lower frequencies. They are robust and cheap and are used in many electrical circuits. The low power consumption and low power dissipation make this device an ideal fit for integrated circuits.

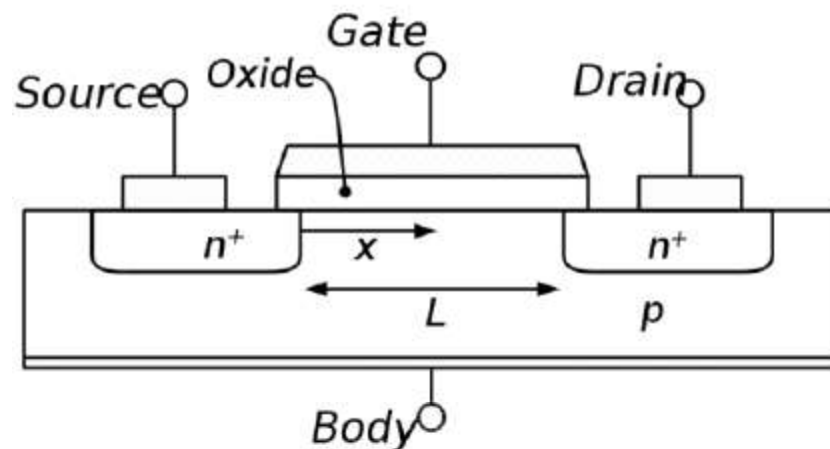


Fig. 4.1 Construction of FET

4.1.1 Working

The FET is an electronic device that contains charge carriers, either electrons or holes that flow from source to drain terminals through the active channel. The conductivity process is controlled by applying the input voltage at the gate terminal.

The current-carrying path that exists between the source and drain terminals is known as “channel” which can be composed of either N-type or P-type semiconductor material. Refer to Figure 4.2 for working of FET.

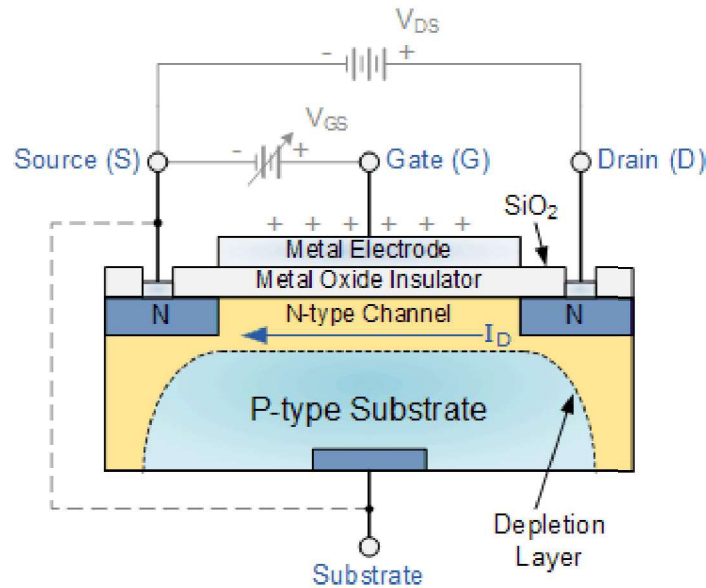


Fig. 4.2 Working of FET

4.1.2 FET Datasheet Specifications & Parameters

FET datasheets contain different parameters and specifications which define the performance of the particular FET type. When developing a new circuit or replacing an existing FET it is important to understand the different parameters and specifications that appear in the datasheets so that the correct device can be chosen and used. All the specifications and parameters are important in different applications.

Interesting Information

The concept of a field-effect transistor (FET) was first patented by Austro-Hungarian physicist Julius Edgar Lilienfeld in 1925 and by Oskar Heil in 1934, but they were unable to build a working practical semiconducting device based on the concept.

Some of the main specifications of FET are defined below. These specifications are always provided in datasheets. Some of the parameters are particularly important for

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different types of FET, e.g., JFET, while others may be more applicable to the MOSFETs etc.

- **Gate source voltage, V_{GS} :** The FET parameter V_{GS} is the rating for the maximum voltage that can be tolerated between the gate and source terminals. The purpose for including this parameter in the datasheet is to prevent damage of the gate oxide. The actual gate oxide withstand voltage is typically much higher than this but it varies as a result of the tolerances that exist in the manufacturing processes. It is advisable to remain well within this rating so that the reliability of the device is maintained.
- **Drain-Source Voltage, V_{DSS} :** This is a rating for the maximum drain-source voltage that can be applied without causing avalanche breakdown. The parameter is normally stated for the case where the gate is shorted to the source. While designing a circuit, it is always best to leave a significant margin between the maximum voltage to be experienced and the V_{DSS} specification.
- **Gate Reverse Leakage Current, I_{gss} :** The current which flows in the reverse bias. It is the leakage current between the gate and source at $V_{DS} = 0$ and is defined by applying the maximum rating V_{GSS} between the gate and source.
- **Threshold voltage $V_{GS(TH)}$:** The threshold voltage $V_{GS(TH)}$ is the minimum gate voltage that can form a conducting channel between the source and the drain.
- **Drain current at zero gate voltage, I_{dss} :** This FET parameter is the maximum continuous current the device can carry with the device fully on. Normally it is specified for a particular temperature, typically 25°C . This FET specification is based on the junction-to-case thermal resistance rating and the case temperature. This FET parameter is of particular interest for power MOSFETs and when determining the maximum current parameter, no switching losses are accounted for. Also holding the case at 25°C is not feasible in practice. As a result, the actual switching current should be limited to less than half of the I_{dss} at $TC = 25^{\circ}\text{C}$ rating in a hard switched application.

- **Gate source cut-off voltage, $V_{GS(off)}$:** The gate source cut-off voltage is a turn-off specification. It defines the threshold voltage for a given residual current, so the device is basically off but on the edge of turning on. The threshold voltage has a negative temperature coefficient, i.e., it decreases with increasing temperature. This temperature coefficient also affects turn-on and turn-off delay times.
- **Forward transconductance, G_{fs} :** Forward transconductance (G_{fs}) represents the signal gain (drain current divided by gate voltage) of a MOSFET.
- **Input capacitance, C_{iss} :** The input capacitance parameter for a FET is the capacitance that is measured between the gate and source terminals with the drain shorted to the source for AC signals. In other words, this is the capacitance between the gate and channel. C_{iss} is made up of the gate to drain capacitance C_{gd} in parallel with the gate to source capacitance C_{gs} . This can be expressed as:

$$C_{iss} = C_{gs} + C_{gd}$$

- **Drain-source on resistance, $R_{ds(on)}$:** With the FET turned hard on, this is the resistance in ohms exhibited across the channel between the drain and source. It is particularly important in switching applications from logic to power switching as well as in RF switching (in radios and televisions), including applications in mixers. FETs typically are able to provide a good performance for switching and have a relatively low $R_{ds(on)}$ value.

FET datasheets contain different parameters and specifications to define the performance of the FET. These are all provided in datasheets that enable the correct choice of FET to be made.

4.2 JFET

The junction-gate field-effect transistor (JFET) is one of the simplest types of field-effect transistor. Figure 4.3 shows the basic form of construction of a practical n-channel and p-channel JFET. A p-channel JFET can be made by transposing the p and n materials as shown in Figure 4.3. All JFETs operate in the depletion mode.

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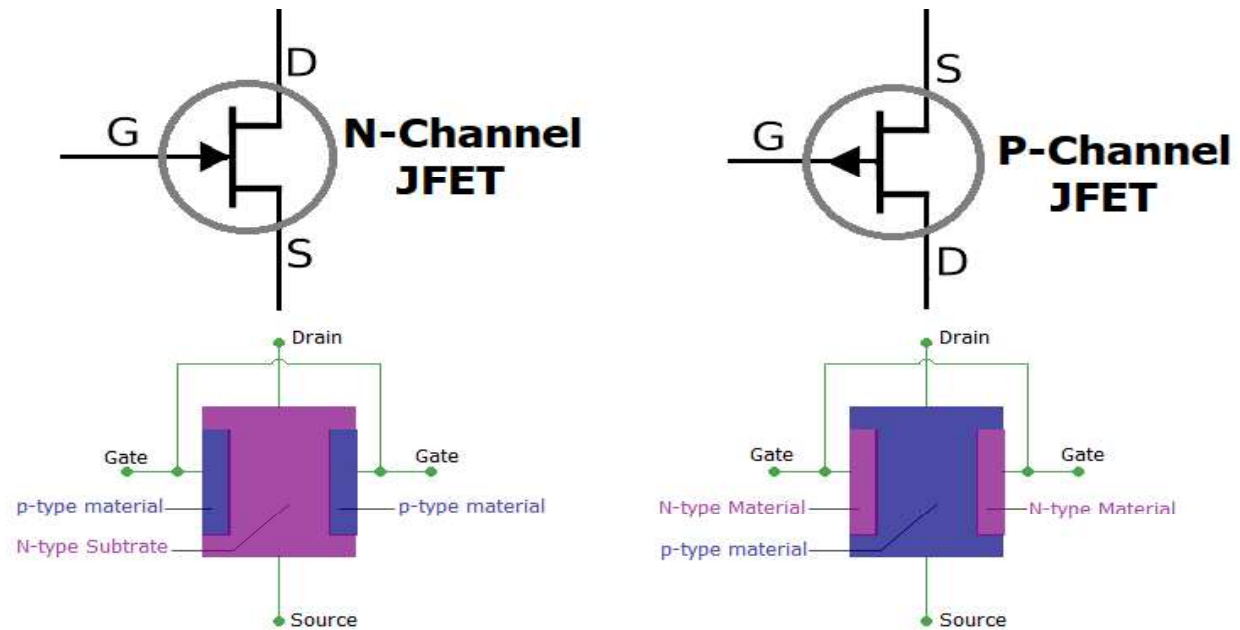


Fig. 4.3 Construction of JFET

Figure 4.4 shows typical transfer characteristics of a low-power n-channel JFET, and illustrates some important features of this type of device. These dopings make a P-N junction in the middle part of the block. We can also form a P-N junction by joining a p-type semiconductor and n-type semiconductor together with a special fabrication technique. The terminal connected to the p-type is the anode. The terminal connected to the n-type side is the cathode.

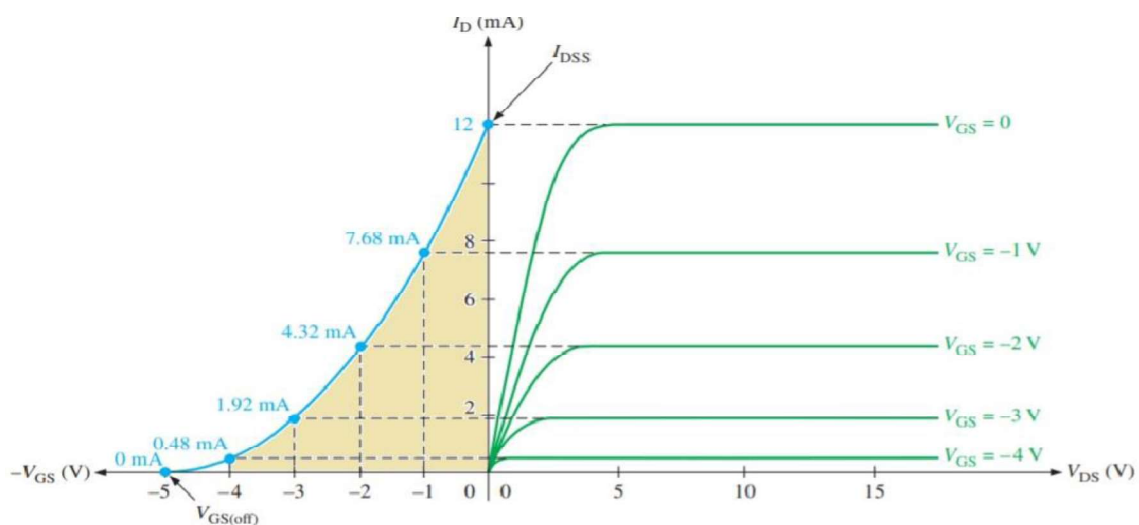


Fig. 4.4 Transfer Characteristics of JFET

The most important characteristics of the JFET are as follows:

- When a JFET is connected to a supply with the polarity shown in Figure 4.2 (drain +ve for an n-channel FET, -ve for a p-channel FET), a drain current (I_D) flows and can be controlled via a gate-to-source bias voltage V_{GS} .
- I_D is maximum when $V_{GS} = 0$, and is reduced by applying a reverse bias to the gate (negative bias in an n-channel device, positive bias in a p-type). The magnitude of V_{GS} needed to reduce I_D to zero is called the ‘pinch-off’ voltage, V_P , and typically has a value between 2 and 10 volts. The magnitude of I_D when $V_{GS} = 0$ is denoted I_{DSS} , and typically has a value in the range 2 to 20mA.
- The JFET’s gate-to-source junction has the characteristics of a silicon diode. When reverse-biased, gate leakage currents (I_{GSS}) are only a couple of nA ($1\text{nA} = .001\mu\text{A}$) at room temperature. Actual gate signal currents are only a fraction of annA, and the input impedance of the gate is typically thousands of mega ohms at lower frequencies.
- If the JFET’s gate-to-source junction is forward-biased, it conducts like a normal silicon diode. If it is excessively reverse-biased, it avalanches like a Zener diode. In both cases, the JFET suffers no damage if gate currents are limited to a few milli amperes.
- Note in Figure 4.4 that, for each V_{GS} value, drain current I_D rises linearly from zero as the drain-to-source voltage (V_{DS}) is increased from zero up to some value at which a ‘knee’ occurs on each curve, and that I_D then remains constant as V_{DS} is increased beyond the knee value. Thus, when V_{DS} is below the JFET’s knee value, the drain-to-source terminals act as a resistor.
- R_{DS} can be varied from a few hundred ohms (at $V_{GS} = 0$) to thousands of mega ohms (at $V_{GS} = V_P$), enabling the JFET to be used as a voltage-controlled switch or as an efficient ‘chopper’ that does not suffer from offset-voltage or saturation-voltage problems.

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- Note in Figure 4.4 that when V_{DS} is above the knee value, the I_D value is controlled by the V_{GS} value and is almost independent of V_{DS} , i.e., the JFET acts as a voltage-controlled current generator. The JFET can be used as a fixed-value current generator by either tying the gate to the source, or by applying a fixed negative bias to the gate. Alternatively, it can (when suitably biased) be used as a voltage-to-current signal amplifier.
- FET ‘gain’ is specified as transconductance, g_m , and denotes the magnitude of change of drain current with gate voltage, i.e., a g_m of 5mA/V signifies that a V_{GS} variation of one volt produces a 5mA change in I_D . Note that the form I/V is the inverse of the ohms formula, so g_m measurements are often expressed in ‘mho’ units. Usually, g_m is specified in FET data sheets in terms of mmhos (milli-mhos) or μ mhos (micro-mhos).
- In most practical applications, the JFET is biased into the linear region and used as a voltage amplifier. N-channel JFET can be used as a common source amplifier (corresponding to the bipolar npn common emitter amplifier).

4.3 MOSFET

The second and most important family of FETs are IGFET or MOSFET. In these FETs, the gate terminal is insulated from the semiconductor body by a very thin layer of silicon dioxide, hence the title ‘Insulated Gate Field Effect Transistor,’ or IGFET. The devices generally use a ‘Metal-Oxide Silicon’ semiconductor material in their construction, hence the alternative title of MOSFET.

Figure 4.5 shows the basic construction and the standard symbol of the n-channel depletion-mode FET. It resembles the JFET, except that its gate is fully insulated from the body of the FET (as indicated by the Figure 4.6) but, in fact, operates on a slightly different principle to the JFET.

It has a normally-open n-type channel between drain and source, but the channel width is controlled by the electrostatic field of the gate bias. The channel can be closed by applying suitable negative bias, or can be increased by applying positive bias.

In practice, the FET substrate may be externally available, making a four-terminal device, or may be internally connected to the source, making a three-terminal device. An important point about the IGFET/MOSFET is that it is also available as an enhancement-mode device, in which its conduction channel is normally closed but can be opened by applying forward bias to its gate.

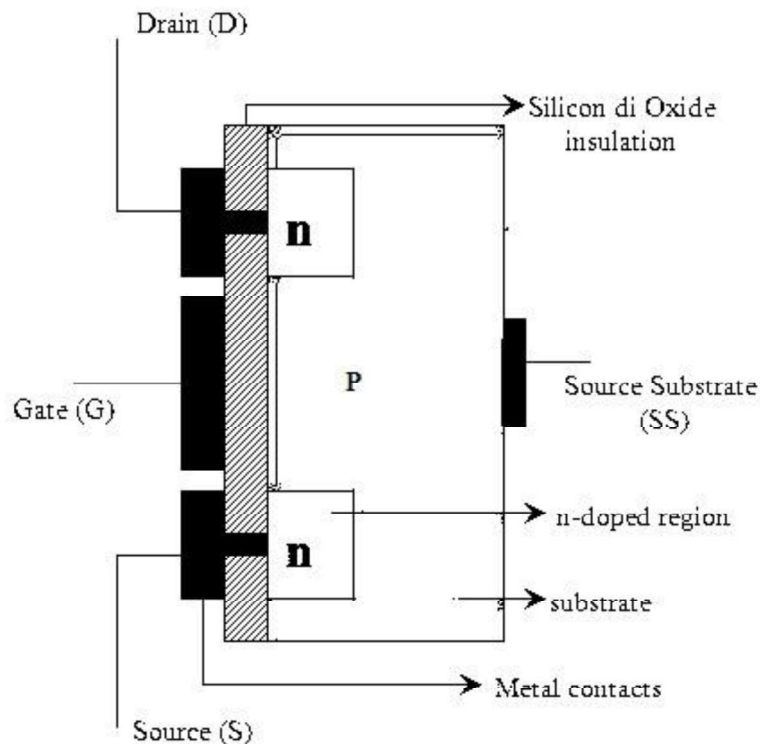


Fig. 4.5 Construction of MOSFET

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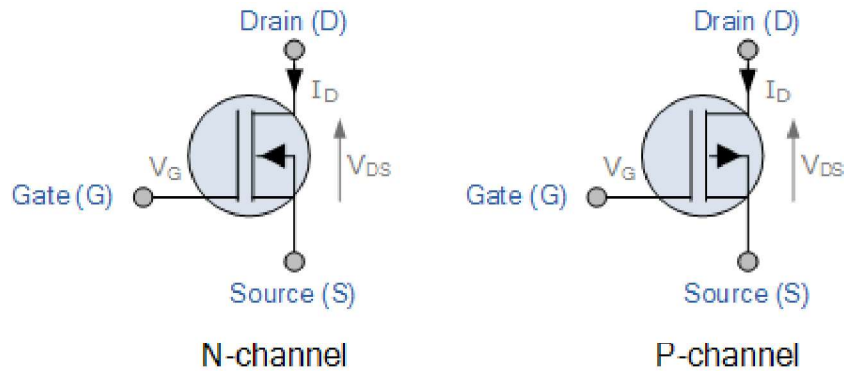


Fig. 4.6 Symbol of MOSFET

To turn the device on, significant positive gate bias is needed, and when this is of sufficient magnitude, it starts to convert the p-type substrate material under the gate into an n-channel, enabling conduction to take place.

Figure 4.7 shows the typical transfer characteristics of an n-channel enhancement-mode IGFET/MOSFET, and the V_{GS}/I_D curves of the same device. Note that no I_D current flows until the gate voltage reaches a ‘threshold’ (V_{TH}) value of a few volts, but that beyond this value, the drain current rises in a non-linear fashion.

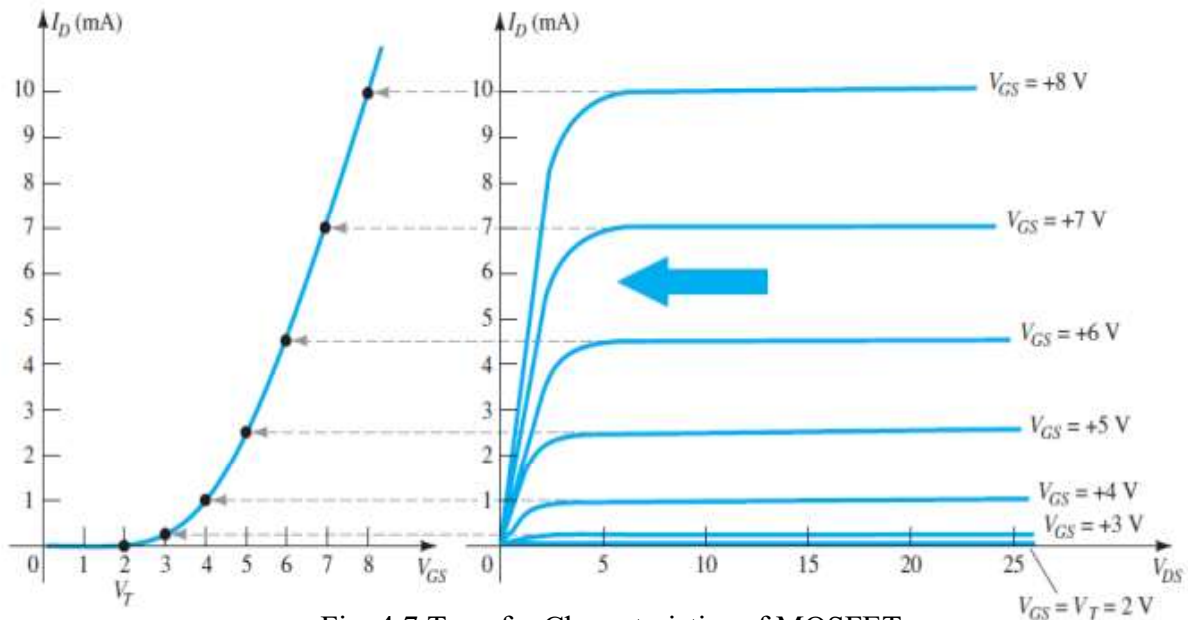


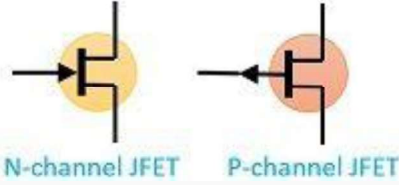
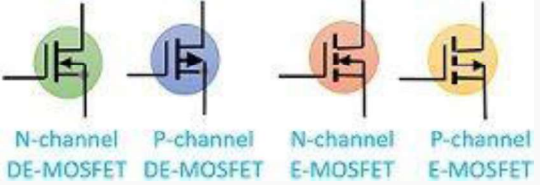
Fig. 4.7 Transfer Characteristics of MOSFET

Note that the transfer graph is divided into two characteristic regions, as indicated (in Figure 4.7) by the dotted line. These are the ‘triode’ region and the ‘saturated’ regions respectively. In the triode region, the device acts like a voltage-controlled resistor. In the saturated region, it acts like a voltage-controlled constant-current generator.

The basic n-channel MOSFETs can be converted to p-channel devices by simply transposing their p and n materials, in which case their symbols must be changed by reversing the directions of their substrate arrows.

Note that the very high gate impedance of MOSFET devices makes them liable to damage from electrostatic discharges and, for this reason, they are often provided with internal protection via integral diodes or Zener diodes.

Comparison Table of JFET and MOSFET

PARAMETERS	JFET	MOSFET
Mode of operation	It operates only in depletion mode.	It can be operated in either depletion or enhancement mode.
Symbol	 N-channel JFET P-channel JFET	 N-channel DE-MOSFET P-channel DE-MOSFET N-channel E-MOSFET P-channel E-MOSFET
Input impedance	JFET have much smaller input impedance mainly of the order of $10^8 \Omega$.	MOSFETs have much higher input impedance of about 10^{10} to $10^{15} \Omega$ due to small leakage current.
Characteristic curve	As JFET has higher drain resistance, the characteristic curve is flatter.	The characteristic curve is less flat than those of JFET.

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PARAMETERS	JFET	MOSFET
Drain resistance	JFET has drain resistance of the order of 10^5 to $10^6 \Omega$	Drain resistance in case of MOSFETs is of the order of 1 to 50 K Ω .
Fabrication	Fabrication process of JFET is more difficult than MOSFET.	MOSFET can be easily fabricated thus it is more widely used.
Cost	Manufacturing of JFET is cheaper as compared to MOSFET.	MOSFETs are slightly expensive as compared to JFET.
Susceptibility to damage	It does not require special handling.	These are more susceptible to overload voltage and requires special handling.

4.4 Thyristors

Thyristors are also switching devices similar to the transistors. Transistors can be used as amplifying and switching device but they cannot handle higher current. So, to handle higher currents, thyristors are used.

Thyristor includes many types of switches, some of them are SCR (Silicon Controlled Rectifier), GTO (Gate Turn OFF), and IGBT (Insulated Gate Controlled Bipolar Transistor) etc. SCR is the most widely used device, so the word Thyristor become synonymous to SCR.

SCR or Thyristor is a four-layered, three-junction semiconductor switching device. It has three terminals: anode, cathode, and gate. Thyristor is also a unidirectional device like a diode, which means it allows current only in one direction. It consists of three P-N junctions in series as it is of four layers. Gate terminal is used to trigger the SCR by providing small voltage to this terminal, which is called gate triggering method to turn ON the SCR.

Difference between Thyristor and MOSFET

Thyristor and MOSFET both are electrical switches and are most commonly used. The basic difference between both of them is that MOSFET switches are voltage-controlled devices and can only switch DC current while Thyristors switches are current controlled device and can switch both DC and AC current. There are some more differences between Thyristor and MOSFET which are given below in the table:

Property	Thyristor	MOSFET
Thermal Run away	Yes	No
Temperature sensitivity	less	high
Type	High voltage high current device	High voltage medium current device
Turning off	Separate switching circuit is required	Not required
Turning On	Single pulse required	No continuous supply is required except during turning On and Off
Switching speed	Low	high
Resistive input impedance	Low	high
Controlling	Current controlled device	Voltage controlled device

4.5 UJT

UJT stands for UniJunction Transistor. It is a three terminal semiconductor switching device. The Unijunction Transistor is a simple device that consists of a bar of n-type

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silicon material with a non-rectifying contact at either end (base 1 and base 2), and with a rectifying contact (emitter) alloyed into the bar part along its length, to form the only junction within the device (hence the name ‘Unijunction’).

The Unijunction Transistor is also known as Double Base Diode. The unique switching characteristics of UJT makes it different from conventional BJT's and FET's by acting as switching transistor instead of amplifying the signals. It exhibits negative resistance in its characteristics which employs it as relaxation oscillators in variety of applications.

4.5.1 Symbol and Construction of Unijunction Transistor (UJT):

In Unijunction Transistor, the P-N Junction is formed by lightly doped N-type silicon bar with heavily doped P-type material on one side. The ohmic contact on either ends of the silicon bar is termed as Base 1 (B_1) and Base 2 (B_2) and P-type terminal is named as emitter.

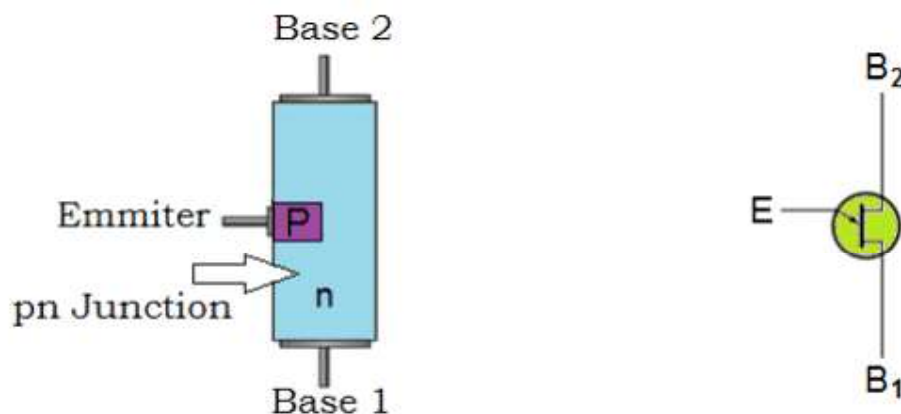


Fig. 4.8 Basic Construction & Symbol of Unijunction Transistor (UJT)

The emitter junction is placed such that it is closer to terminal Base 2 than Base 1. The symbols of both UJT and JFET resemble the same except the emitter arrowhead represents the direction in which conventional current flow, but they operate differently.

4.5.2 Working of a Unijunction Transistor

The equivalent circuit (Figure 4.9) shows that N-type channel consists of two resistors R_{B2} and R_{B1} in series with an equivalent diode, D representing the PN junction. The emitter PN junction is fixed along the ohmic channel during its manufacturing process.

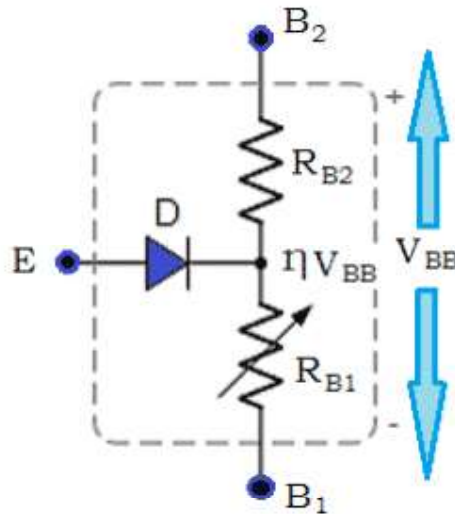


Fig. 4.9 Simplified Equivalent Circuit of Unijunction Transistor

The variable resistance R_{B1} is provided between the terminals Emitter (E) and Base 1 (B_1), the R_{B2} between the terminals Emitter (E) and Base 2 (B_2). Since the PN junction is closer to B_2 , the value of R_{B2} will be less than the variable resistance R_{B1} .

A voltage divider network is formed by the series resistances R_{B2} and R_{B1} . When a voltage is applied across the semiconductor device, the potential will be in proportion to the position of base points along the channel. The Emitter (E) will act as input when employed in a circuit, as the terminal B_1 will be grounded. The terminal B_2 will be positive biased to B_1 , when a voltage (V_{BB}) applied across the terminals B_1 and B_2 . When the emitter input is zero, the voltage across resistance R_{B1} of the voltage divider circuit is calculated by:

$$V_{RB_1} = \frac{R_{B1}}{R_{B1} + R_{B2}} (V_{BB})$$

The important parameter of Unijunction Transistor is 'intrinsic stand-off ratio' (η), which is resistive ratio of R_{B1} to R_{BB} . Most UJT's have η value ranging from 0.5 to 0.8. The P-N

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junction is reverse biased; when small amount of voltage which is less than voltage developed across resistance R_{B1} (ηV_{BB}) is applied across the terminal emitter (E). Thus, a very high impedance is developed prompting device to move into non-conducting state i.e., it will be switched off and no current flows through it. The UJT begins to conduct when the P-N junction is forward biased.

The forward biased is achieved when voltage applied across emitter terminal is increased and becomes more than V_{RB1} . This results in larger flow of emitter current from emitter region to base region. Increase in emitter current reduces the resistance between emitter and Base 1, resulting in negative resistance at emitter terminal.

The Unijunction Transistor (UJT) will act as voltage breakdown device, when the input applied between emitter and base 1 reduces below breakdown value i.e., R_{B1} increases to a higher value. This shows that R_{B1} depends on the emitter current and it is variable.

4.5.3 Characteristics Curve of Unijunction Transistor (UJT)

The characteristics of Unijunction Transistor (UJT) can be explained by three parameters:

- Cutoff
- Negative Resistance Region
- Saturation

Figure 4.10 shows the detailed characteristics of UJT.

Cutoff

Cutoff region is the area where the Unijunction Transistor (UJT) doesn't get sufficient voltage to turn on. The applied voltage hasn't reached the triggering voltage, thus making transistor to be in off state.

Negative Resistance Region

When the transistor reaches the triggering voltage, V_{TRIG} , Unijunction Transistor (UJT) will turn on. After a certain time, if the applied voltage increases to the emitter lead, it

will reach out at V_{PEAK} . The voltage drops from V_{PEAK} to Valley Point even though the current increases (negative resistance).

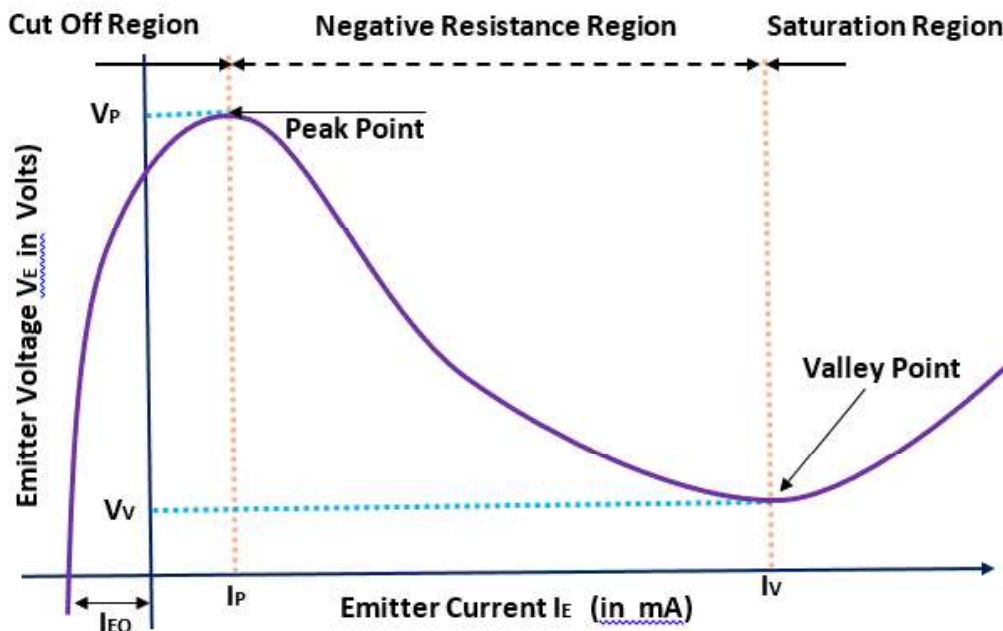


Fig. 4.9 Characteristics of Unijunction Transistor (UJT)

Saturation

In saturation region, current increases with the increase in the applied voltage to emitter terminal.

4.5.4 Applications of Unijunction Transistor (UJT)

The Unijunction Transistor can be employed in variety of applications such as:

- Switching Device
- Triggering Device for Triacs and SCR's
- Timing Circuits
- For phase control
- In sawtooth generators
- In simple relaxation oscillators

Activity

- Implement UJT as a switch on breadboard.

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Advantages of Unijunction Transistor (UJT)

The advantages of Unijunction Transistor include

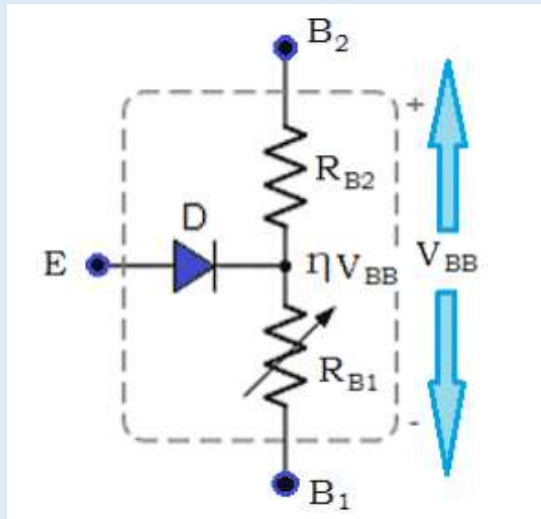
- Low cost
- Negative resistance characteristics
- Requires low value of triggering current.
- A stable triggering voltage
- Low power absorbing device

Disadvantage of Uni junction Transistor (UJT)

The main disadvantage of Unijunction Transistor is its inability to provide appropriate amplification

Example 4.1

The intrinsic stand-off ratio for a UJT is determined to be 0.6. If the inter-base resistance (R_{BB}) is $10\text{k}\Omega$ what are the values of R_{B1} and R_{B2} ?



Solution:

Intrinsic stand-off ratio for a UJT is given as

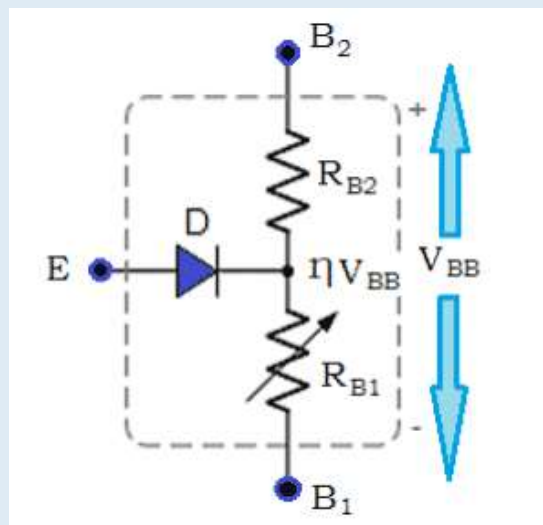
$$\eta = \frac{R_{B1}}{R_{B1} + R_{B2}}$$
$$0.6 = \frac{R_{B1}}{10K}$$
$$R_{B1} = 6K\Omega$$

Teacher Notes

All the topics of this chapter should be taught covering only the basics. Details should be as minimum as possible.

Example 4.2

A UJT has 10V between the bases. If the intrinsic stand-off ratio is 0.65, find the value of stand-off voltage. What will be the peak point voltage if the forward voltage drop in the p-n junction is .7V?



Solution:

$$V_{BB}=10V, \eta=0.65, V_D=0.7V$$

Stand-off voltage (V_{RB1}) is given as

$$V_{RB1} = \eta V_{BB}$$

$$V_{RB1} = 0.65 \times 10$$

$$V_{RB1} = 6.5V$$

Peak point Voltage (V_P) is given as

$$V_P = \eta V_{BB} + V_D$$

After putting values and solving:

$$V_P = 7.2V$$

Key Points

- The FET (Field Effect Transistor) is a three-terminal electronic device used to control the flow of current by the voltage applied to its gate terminal.
- The conductivity process in FET is controlled by applying the input voltage at the gate terminal.
- The junction-gate field-effect transistor (JFET) is one of the simplest types of field-effect transistor.
- All JFETs operate in the depletion mode.
- In MOSFETs, the gate terminal is insulated from the semiconductor body by a very thin layer of silicon dioxide.
- In the triode region, MOSFET acts like a voltage-controlled resistor; in the saturated region, it acts like a voltage-controlled constant-current generator.
- Thyristors are also switching devices similar to the transistors and they are capable of operating in high current conditions.
- The unique switching characteristics of UJT makes it different from conventional BJT's and FET's by acting as switching transistor instead of amplifying the signals.
- The main disadvantage of Unijunction Transistor is its inability to provide appropriate amplification.

Exercise

Select the most appropriate option

- Which of the following are three terminal electronic devices:
a. Resistance b. FET c. MOSFET d. Both b and c
- In MOSFETs, which terminal is insulated from the semiconductor body.
a. Drain b. Source c. Gate d. None of these
- _____ are like transistor but can operate in high current conditions.
a. BJT b. Thyristors c. MOSFETs d. None of these
- _____ is also known as a double base diode.
a. UJT b. BJT c. Zener d. JFET
- _____ are voltage controlled devices
a. JFET b. MOSFET c. Thyristor d. None of these

Write short answer of the following.

- Define FET.
- Define MOSFET?
- What is the difference between thyristors and MOSFETs.
- Define UJT.
- What is the disadvantage of using UJT?

Answer the following question in detail.

- Explain the working of MOSFET.
- Differentiate between UJT and Thyristor.
- Describe the working of UJT.
- Enlist Applications of UJT.
- List down advantages of UJT.

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Solve the Following Problems:

1. The intrinsic stand-off ratio for a UJT is determined to be 0.5. If the inter-base resistance (R_{BB}) is $12\text{k}\Omega$ what are the values of R_{B1} and R_{B2} ?
2. A UJT has 15V between the bases. If the intrinsic stand-off ratio is 0.75, find the value of stand-off voltage. What will be the peak point voltage if the forward voltage drop in the p-n junction is .7V?
3. Determine the minimum and maximum peak-point voltage for UJT with $V_{BB}=24\text{V}$. Given that UJT has a range of $\eta=0.74$ to 0.86.